

Learning Discriminative Representations to Interpret Image Recognition Models

Thèse de Doctorat

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- 2 Opti-CAM: Optimizing saliency maps for interpretability
- 3 CA-Stream: Attention-based pooling for interpretable image recognition
- 4 A learning paradigm for interpretable gradients
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Motivation

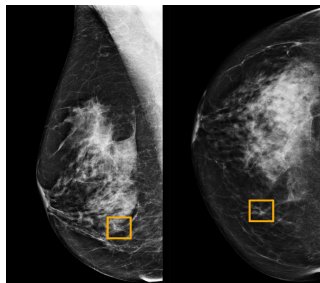
High stakes

Consider these situations:

Motivation

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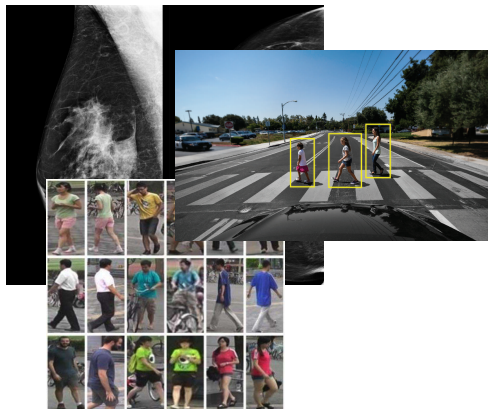
Consider these situations:



Motivation

High stakes

Consider these situations:



Straight to the point



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

We must **understand** the
behaviour of these models.

Step by step

Computation, Computer Vision and AI



Explainable AI



Thesis objectives

Computation



Better known as *Computer Science*.

Study of:

- Algorithms.
- Data structures.
- Design of hardware and software.

Computation, Computer Vision and AI

Computer Vision

Replication of human vision capabilities.

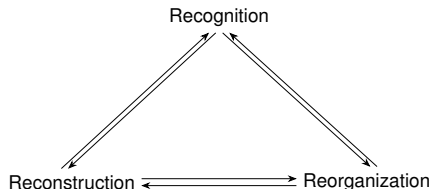
Computation, Computer Vision and AI

Computer Vision

Replication of human vision capabilities.

Three fundamental tasks[1]:

- Recognition.
- Reconstruction.
- Reorganization.



Computation, Computer Vision and AI

Artificial Intelligence

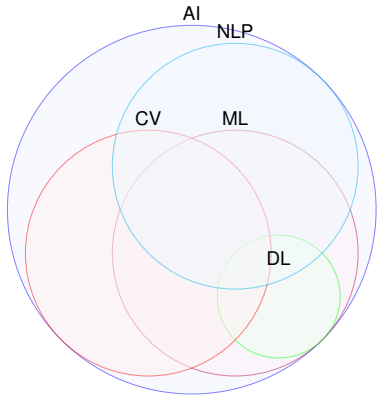
Systems capable of performing tasks requiring human intelligence [2].

Computation, Computer Vision and AI

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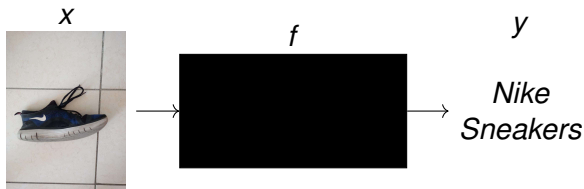
Subfields:

- Machine Learning (*ML*) & Deep Learning (*DL*).
- Computer Vision(*CV*).
- Natural Language Processing (*NLP*).
- Robotics.



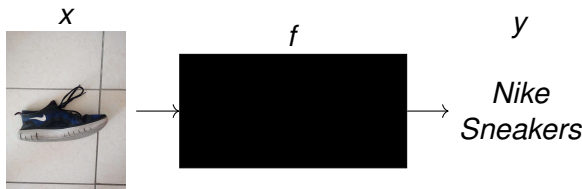
Explainable AI

We are interested in understanding models, behaving like a black box model:



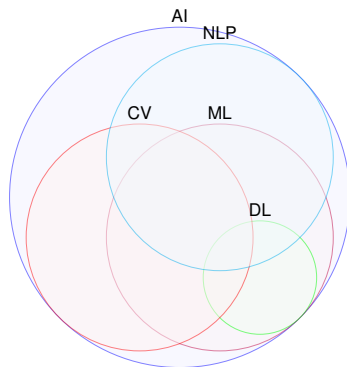
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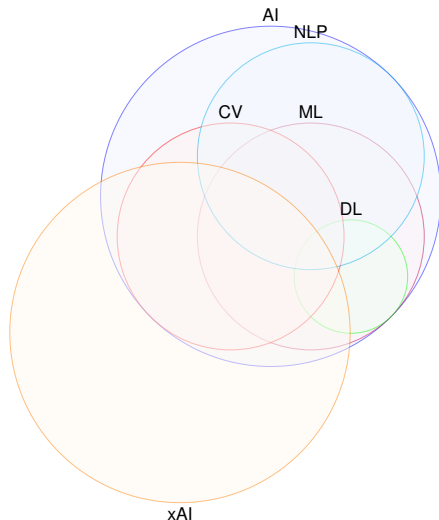


We want to *know why* $f(x) \rightarrow y$

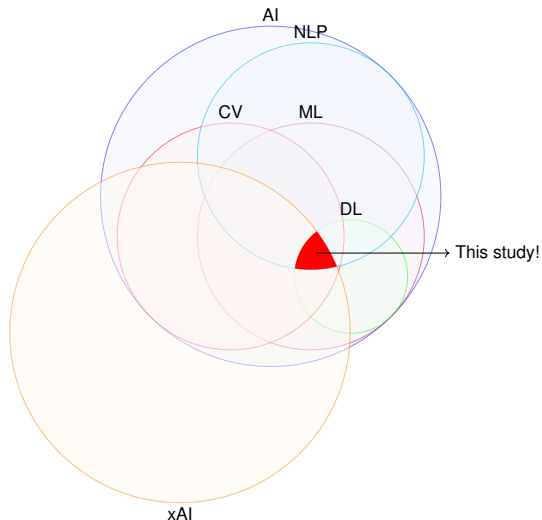
Fitting it all together



Fitting it all together



Fitting it all together



Thesis Objectives

Improvement of recognition and interpretable properties of model predictions.

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Improvement of recognition and interpretable properties of model predictions.

In particular:

- Development of low cost/complexity explainability approaches.
- Establishment of a fixed evaluation protocol.
- Differentiation of human based and machine explanations.

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Preliminaries

Desired data of Interpretability Study [4]

We ask questions
regarding *black box*
models.

- How does it work?
- How safe is it?
- Who is accountable for it?
- Who benefits from it?
- Who uses it?

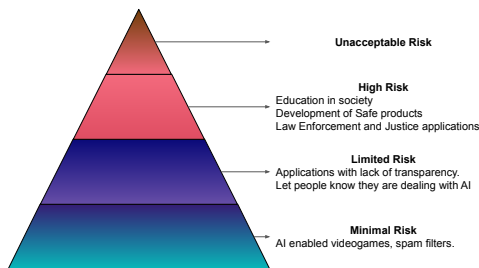
**Accepted AI systems
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Preliminaries

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Regulation planned with the European AI act[3]

Accepted AI systems within society must answer this.

Image Recognition Models

Convolutional Neural Networks (CNN)

Based on the **convolution operation**.

A representation $f \star g$ is computed for a feature map f and a kernel g .

First approach with *Neocognitron*

[5]

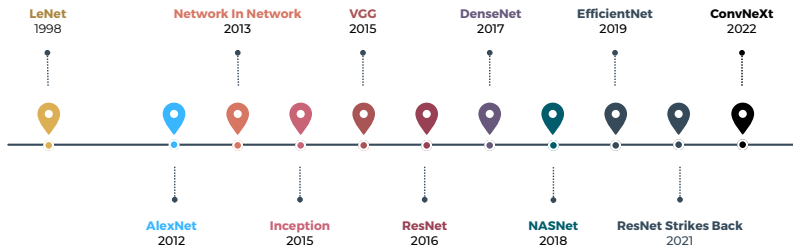
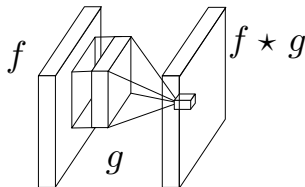


Image Recognition Models

Convolutional Neural Networks(CNN)

Training deep models is difficult, gradient information *is* lost with depth.

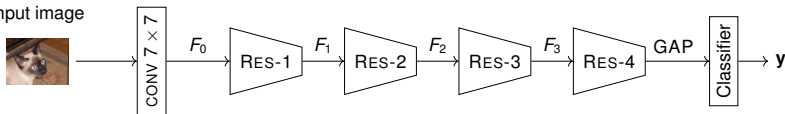
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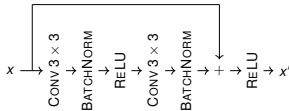
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A model surviving the test of time: *ResNet* [6].

Input image



Basic Block



Bottleneck Block

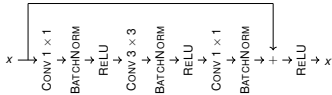


Image Recognition Models

Self Attention Architectures

Transformers

Attention: compute the relevance of each element relative to others in an embedding.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{Q \times K^T}{\sqrt{d_k}} \right) V$$

Image Recognition Models

Self Attention Architectures

A modification of transformers to vision, the *Vision Transformer* (ViT)
[8]

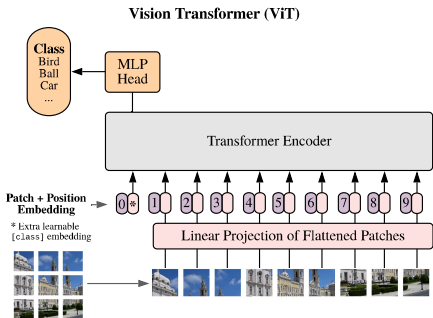


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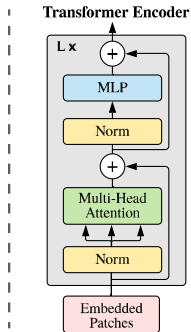
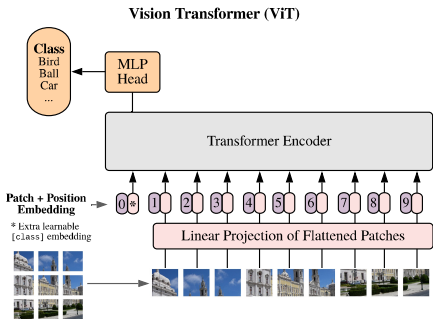
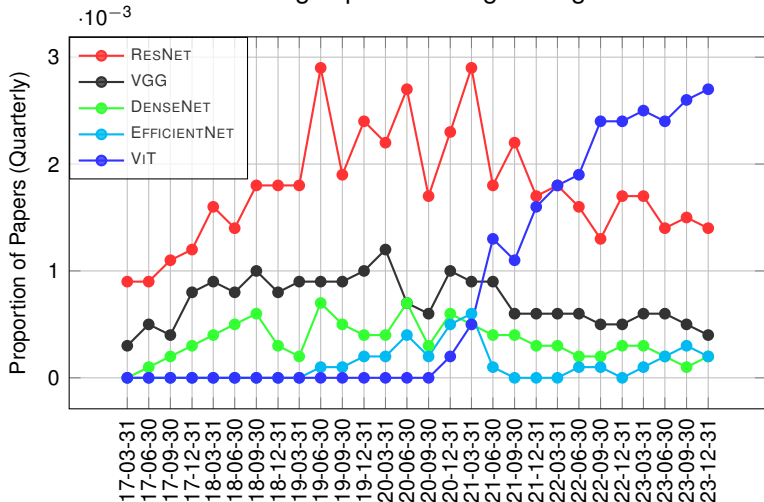


Image Recognition Models

The Current Landscape

Transformers had a strong impact on image recognition:



Interpretability

Interpretability

Two dimension grouping

Transparency

Approaches requiring changes to an architecture or training process.

Post-Hoc Interpretability

Approaches utilizing information contained within a model.

Interpretability

Class Activation Methods

Semantic information is found in deeper stages of a model [9].



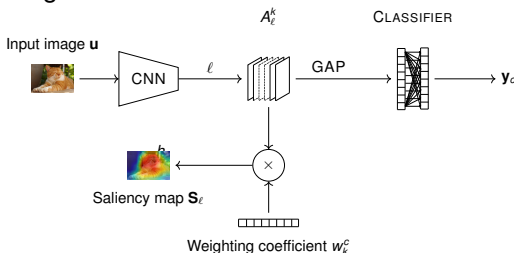
Interpretability

Class Activation Methods

Semantic information is found in deeper stages of a model [9].



Salient regions are highlighted for an input image:



Governed by equation:

$$S_\ell^c(\mathbf{u}) := h \left(\sum_k w_k^c A_\ell^k \right) \quad (1)$$

Interpretability

Evaluating Interpretability

Saliency: what is relevant towards a prediction.

Image Classification Approach

An explanation should demonstrate similar predictive properties to its query:

Input image. (I)



$$P_i^c = 0.8756$$

Explanation Map. (E^c)



$$O_i^c = 0.7442$$

Interpretability

Evaluating Interpretability

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Image Classification Approach

An explanation should demonstrate similar predictive properties to its query:

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$$P_i^c = 0.8756$$

- Prediction loss when using an explanation. (Average Drop (AD) [10])
- Ratio of explanations being better than the input image. (Average Increase (AI) [10])
- Reaction to saliency guided modifications (Insertion/Deletion (Ins/Del) [11])

Explanation Map. (E^c)



$$O_i^c = 0.7442$$

Interpretability

Evaluating Interpretability

Saliency: what is relevant towards a prediction.

Weakly Supervised Object Localization Approach

A saliency map should be good to localize the object of interest for an example:

Input image. (I)



Saliency Map. (E^c)



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A saliency map should be good to localize the object of interest for an example:

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Saliency Map. (E^c)



- Measurements between the predicted object bounding box and object bounding boxes (OM, LE, F_1).
- Overlapping between largest connected bounding box and object bounding box (BA).
- Existence of the most salient points and the fraction of the map with the bounding box (EP, SP).
- Penalty of size of the bounding box with entropy and image size (SM).

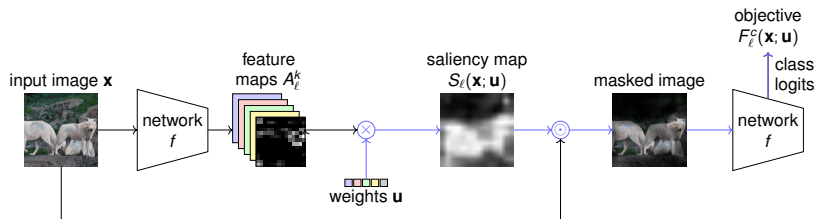
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Preliminaries

Premise

Is it possible to optimize the weighting coefficient w_k^c to obtain a saliency map with the highest probability of prediction?



Preliminaries

Contributions

Opti-CAM

Maximization of predicted probability of an explanation.

Average Drop

Evaluation metric designed to enhance comparisons.

Opti-CAM

Motivation

- **Combination of Feature Maps**

To compute w_k^c , Score-CAM [12] considers feature maps independently.

What about a linear combination?

$$F(\mathbf{w}) := f \left(\mathbf{u} \odot n \left(\text{up} \left(\sum_k w_k A_l^k \right) \right) \right)_c$$

F is optimizable with respect to $\mathbf{w}$ and an initial w_0 via backpropagation.

Opti-CAM

Motivation

● Combination of Feature Maps

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F is optimizable with respect to $\mathbf{w}$ and an initial w_0 via backpropagation.

● Masking-based Saliency Saliency maps produced by optimization in the input space [13, 14]

The optimization usually takes form of:

$$S^c(\mathbf{u}) := \underset{\mathbf{m} \in M}{\operatorname{argmax}} f(\mathbf{u} \odot n(\text{up}(\mathbf{m})))_c + \delta R(\mathbf{m})$$

Opti-CAM

Optimization

Formulation Assuming channel weights $w_k := \text{softmax}(\mathbf{u})_k$, the saliency map S_ℓ is a function of both an input image x and a variable $\mathbf{u}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (2)$$

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Find vector $\mathbf{u}^*$ maximizing confidence for class c , with an input $\mathbf{x}$ masked by Saliency Map $S_\ell(\mathbf{x}; \mathbf{u}^*)$:

$$\mathbf{u}^* := \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}^*), \quad (3)$$

The objective function:

$$F_\ell^c(\mathbf{x}; \mathbf{u}) := g_c(f(\mathbf{x} \odot S_\ell(\mathbf{x}; \mathbf{u}))) \quad (4)$$

Opti-CAM

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The Saliency Map then becomes:

$$S_\ell^c(\mathbf{x}) := S_\ell(\mathbf{x}; \mathbf{u}^*) = S_\ell(\mathbf{x}; \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u})) \quad (5)$$

Average Gain

Motivation

Existing metrics for measuring interpretability ($AD\downarrow$, $AI\uparrow$), can be fooled.

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Existing metrics for measuring interpretability ($AD \downarrow$, $AI \uparrow$), can be fooled.

Fake-CAM [15]

What happens if you provide a Saliency Map where all the pixels *but-one* are 1?

Input



$$p_{389} = 0.874$$

Explanation



$$o_{389} = 0.822$$

● Drop = 5.95%

● Increase = 0%

Therefore, we don't drop much predictive power, *but*, we don't gain confidence in our prediction.

Average Gain

Definition

With the notation on slide (22):

$$AG(\%) := \frac{1}{N} \sum_{i=1}^N \frac{[o_i^c - p_i^c]_+}{1 - p_i^c} \cdot 100 \quad (6)$$

Measuring the increase in predictive power when masking an input.

Experimental Set-Up

Implementation

- All images are set to $224 \times 224 \times 3$.
- Adam [16] optimizer, for 100 iterations.
- Cross Entropy.
- Stop optimizing if loss is less than 10^{-10} .

¹Pytorch Model Zoo <https://pytorch.org/vision/0.8/models.html>

²Timm <https://github.com/rwightman/pytorch-image-models>

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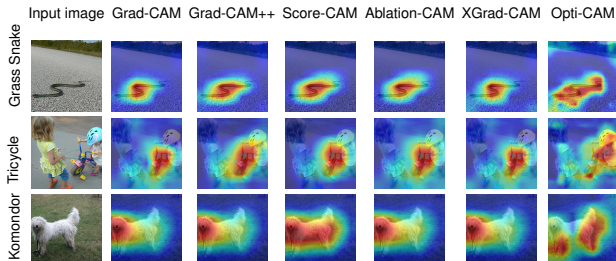
Datasets and Models

- **ILSVRC** Validation splits of ImageNet ([17, 18]), 50k images, over 1k classes.
- **Medical Data** Chest X-ray images with 3 diseases [19], endoscopy images for pathologies, findings and landmarks [20].
- **Networks** ResNet-50 [6], and VGG [21]. ¹ViT-B (16-224) [8] and DeiT-B (16-224) [22] ².

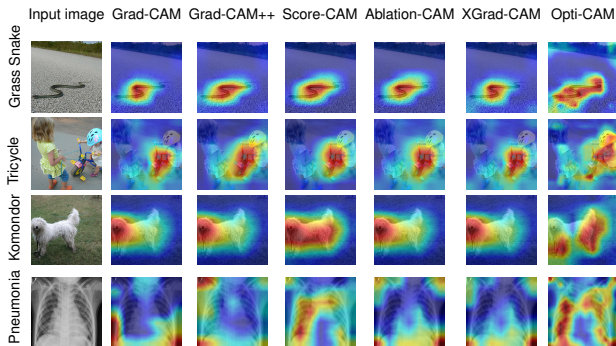
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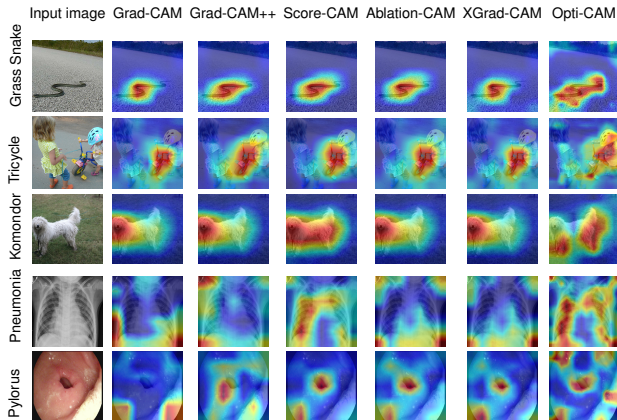
Qualitative Evaluation



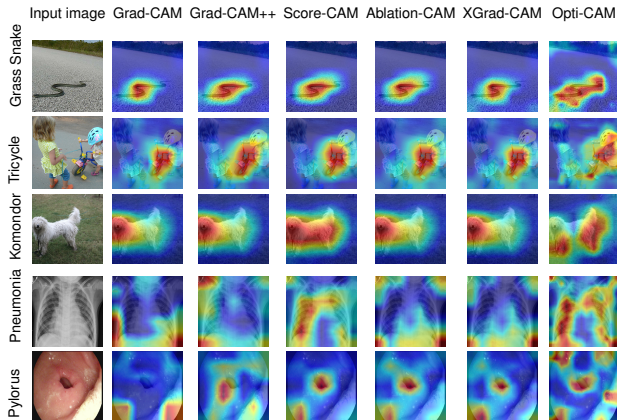
Qualitative Evaluation



Qualitative Evaluation



Qualitative Evaluation



Results

Quantitative Experiments

Classification Metrics on CNNs:

METHOD	RESNET50						VGG16					
	AD↓	AG↑	AI↑	I↑	D↓	T(s)	AD↓	AG↑	AI↑	I↑	D↓	T(s)
Fake-CAM	0.8	1.6	46.0	50.7	28.1	0.00	0.5	0.6	42.6	46.1	26.9	0.00
Grad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	14.2	14.7	40.6	64.1	11.6	0.02
Grad-CAM++	12.9	16.0	42.1	66.0	14.7	0.03	17.1	10.2	33.4	62.9	12.2	0.02
Score-CAM	8.6	26.6	56.7	65.7	16.3	15.22	13.5	15.6	41.7	62.5	12.1	3.11
Ablation-CAM	12.5	16.4	42.8	65.9	14.6	18.26	15.5	12.6	36.9	63.8	11.4	2.98
XGrad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	13.8	14.8	41.2	64.1	11.7	0.02
Layer-CAM	15.6	15.0	38.8	67.0	14.2	0.08	48.9	3.1	13.5	58.3	6.4	0.07
ExPerturbation	38.1	9.5	22.5	70.7	15.0	152.96	43.0	7.1	20.5	61.1	15.0	83.20

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ExPerturbation	38.1	9.5	22.5	70.7	15.0	152.96	43.0	7.1	20.5	61.1	15.0	83.20
Opti-CAM	1.5	68.8	92.8	62.0	19.7	4.15	1.3	71.2	92.7	59.2	11.0	3.94

Results

Localization Experiments

METHOD	RESNET50								VGG16							
	OM↓	LE↓	F1↑	BA↑	SP↑	EP↑	SM↓	OM↓	LE↓	F1↑	BA↑	SP↑	EP↑	SM↓		
Fake-CAM	63.6	54.0	57.7	47.9	99.8	28.5	0.98	64.7	54.0	57.7	47.9	99.8	28.5	1.07		
Grad-CAM	72.9	65.8	49.8	56.2	69.8	33.3	1.30	71.1	62.3	42.0	54.2	64.8	32.0	1.39		
Grad-CAM++	73.1	66.1	50.4	56.2	69.9	33.1	1.29	70.8	61.9	44.3	55.2	66.2	32.3	1.38		
Score-CAM	72.2	64.9	49.6	54.5	68.7	32.4	1.25	71.2	62.5	45.3	58.5	68.2	33.4	1.40		
Ablation-CAM	72.8	65.7	50.2	56.1	69.9	33.1	1.26	71.3	62.6	43.2	56.2	65.7	32.7	1.39		
XGrad-CAM	72.9	65.8	49.8	56.2	69.8	33.3	1.30	70.8	62.0	41.9	53.5	64.4	31.6	1.41		
Layer-CAM	73.1	66.0	50.1	55.5	70.0	33.0	1.29	70.5	61.5	28.0	54.7	65.0	32.4	1.45		
ExPerturbation	73.6	66.6	37.5	44.2	64.8	38.2	1.59	74.1	66.4	37.8	43.3	62.7	36.1	1.74		

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Layer-CAM	73.1	66.0	50.1	55.5	70.0	33.0	1.29		70.5	61.5	28.0	54.7	65.0	32.4	1.45	
ExPerturbation	73.6	66.6	37.5	44.2	64.8	38.2	1.59		74.1	66.4	37.8	43.3	62.7	36.1	1.74	
Opti-CAM	72.2	64.8	47.3	49.2	59.4	30.5	1.34		69.1	59.9	44.1	51.2	61.4	30.7	1.34	

Conclusions

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Preliminaries

Premise

CAM activation methods compute saliency maps in a similar fashion that transformers generate abstract representations via the [CLS] token:

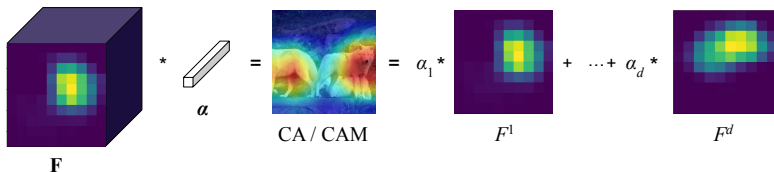


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